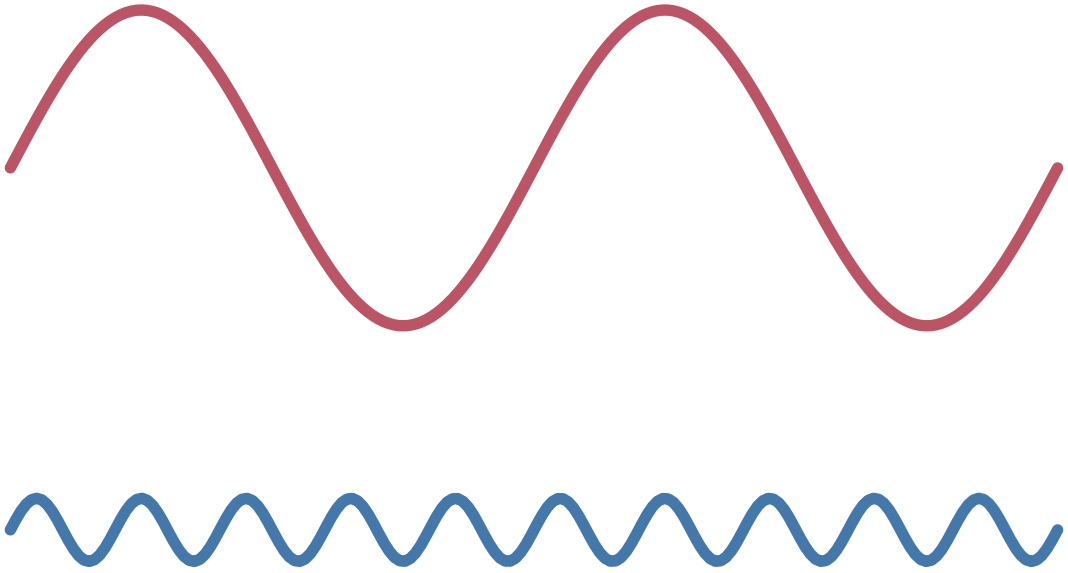
Harmonics & Resonance

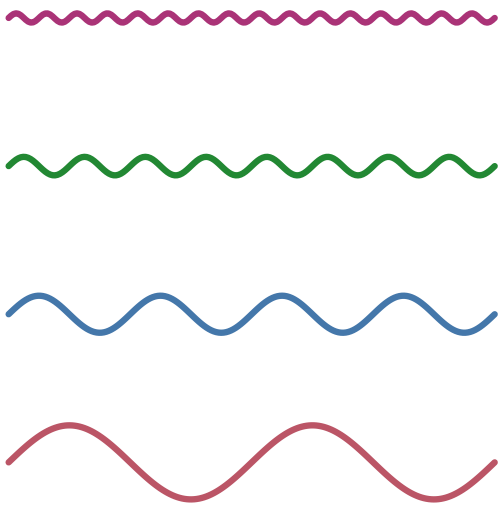
Dr. Josef Fruehwald

Complex Sounds



Complex Sounds

Complex Sounds

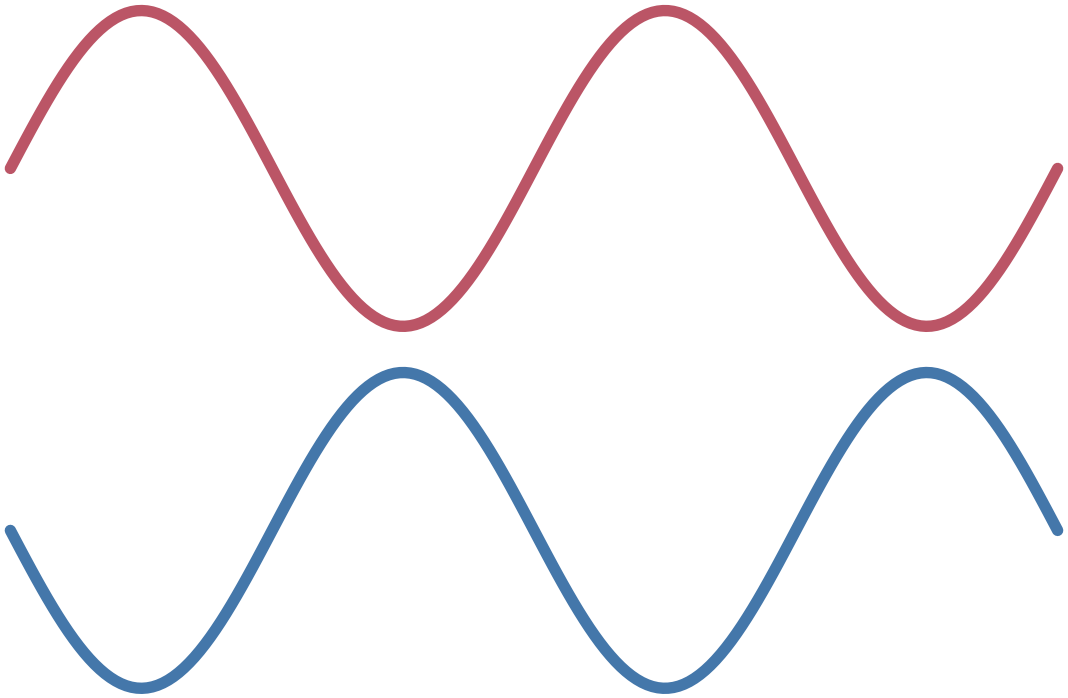




Complex sounds

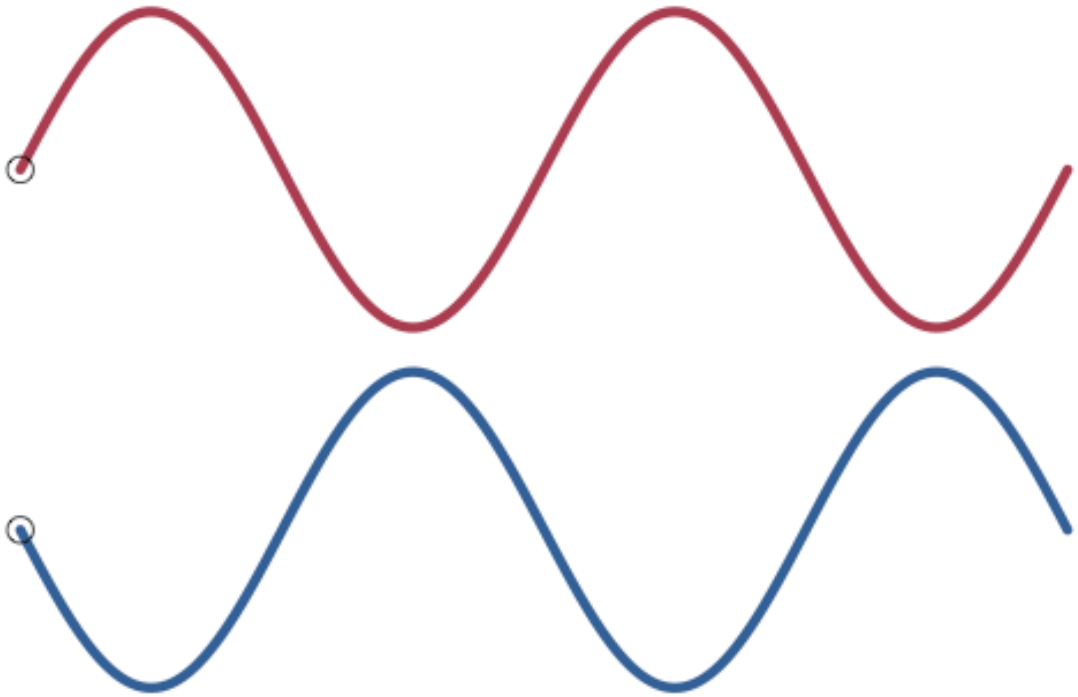
Complex Sounds

Interference

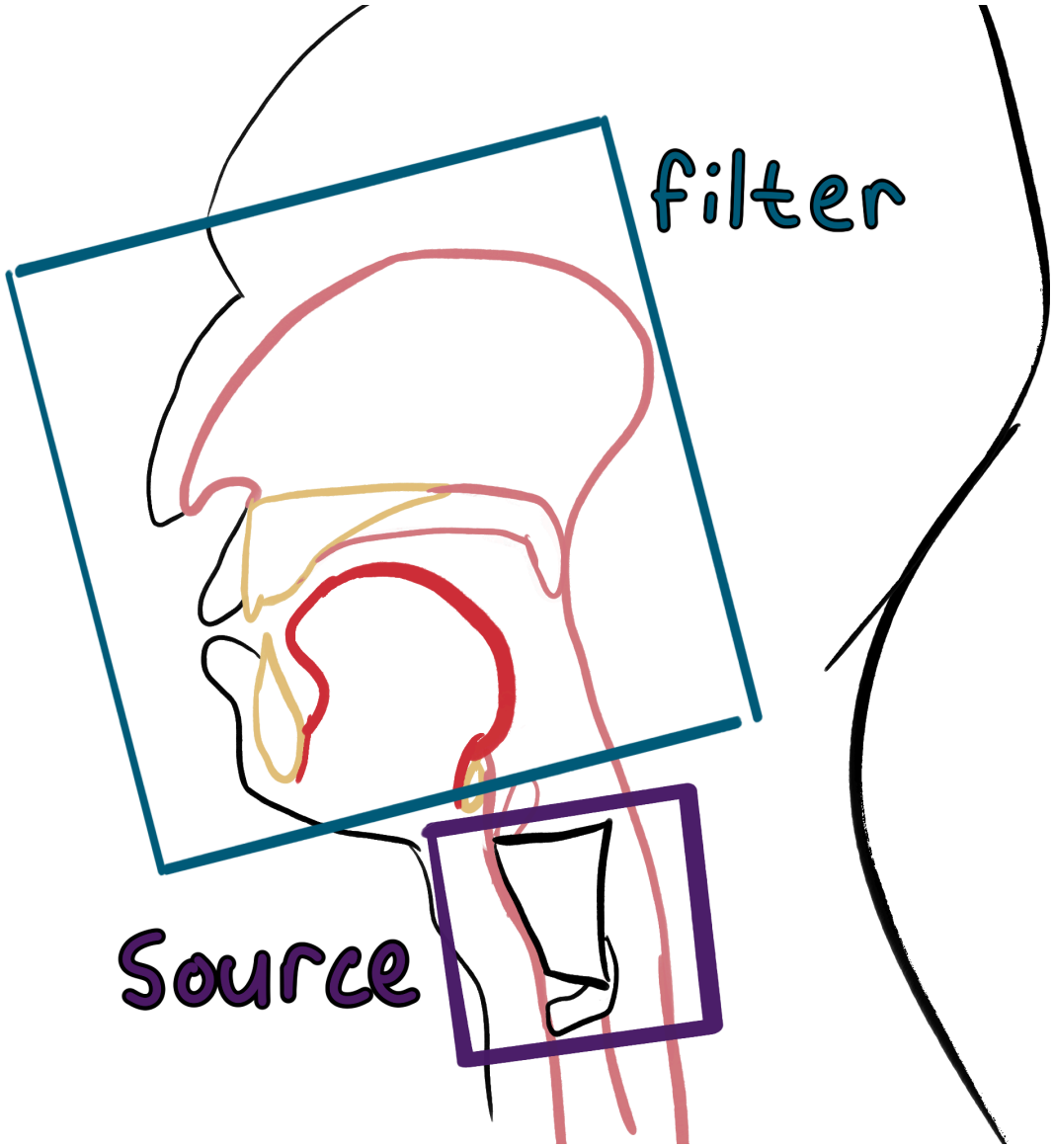


Complex Sounds

Interference

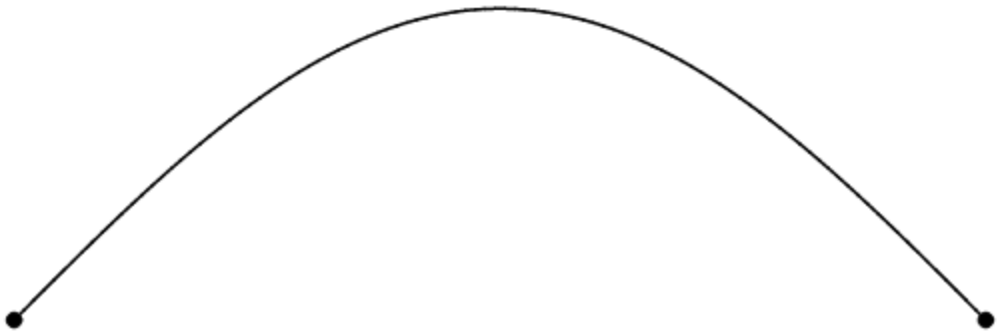


The Source-Filter Model



Fundamental Frequency - From the Source

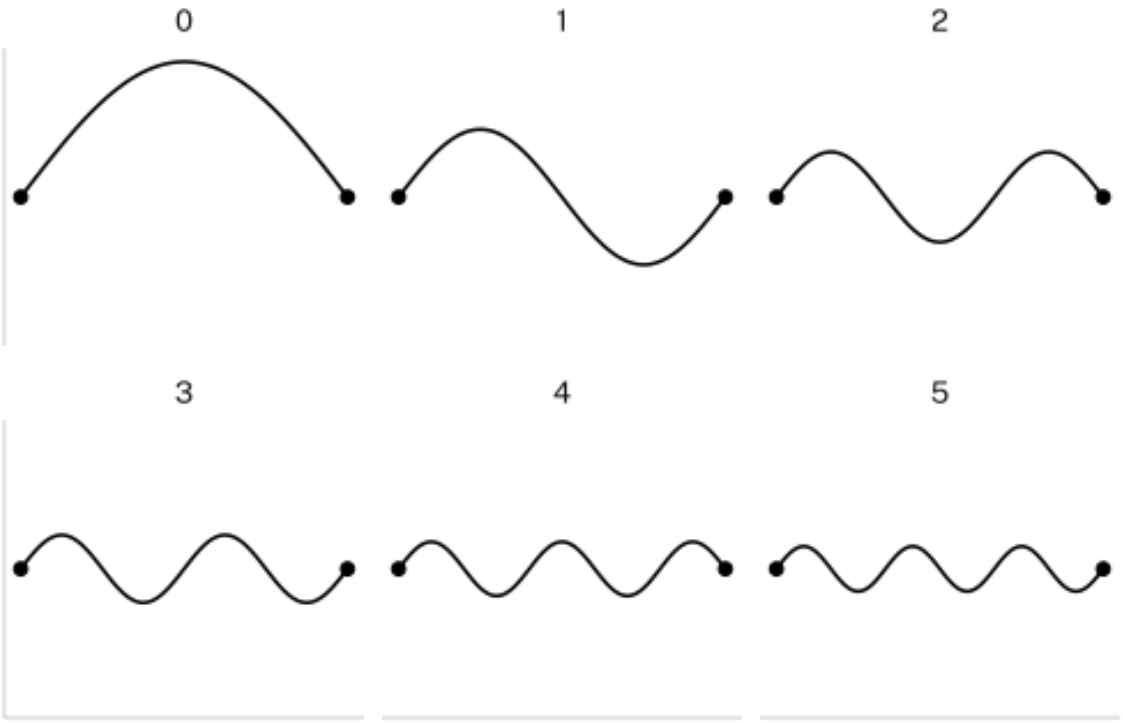
The wavelength is proportional to the string length.



Fundamental Frequency - From the Source

Wave Length = 2Length

Harmonics - From the Source



Harmonics - From the Source

$$WL_n = \frac{2L}{n}$$

- When n = 1, WL is the fundamental frequency.
- For integers > 1, WL is a harmonic.
- Every harmonic wavelength is the fundamental divided by a whole number integer.
- Every harmonic *frequency* is a whole number *multiple* of the fundamental frequency.

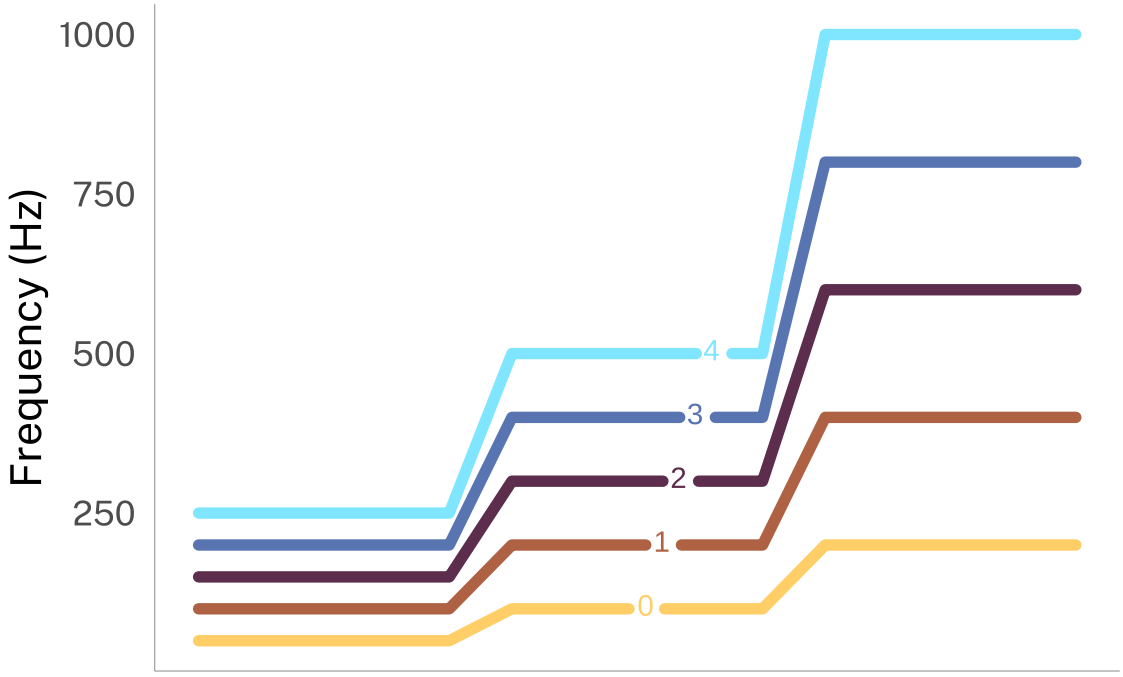
Harmonics

Wavelength to Frequency

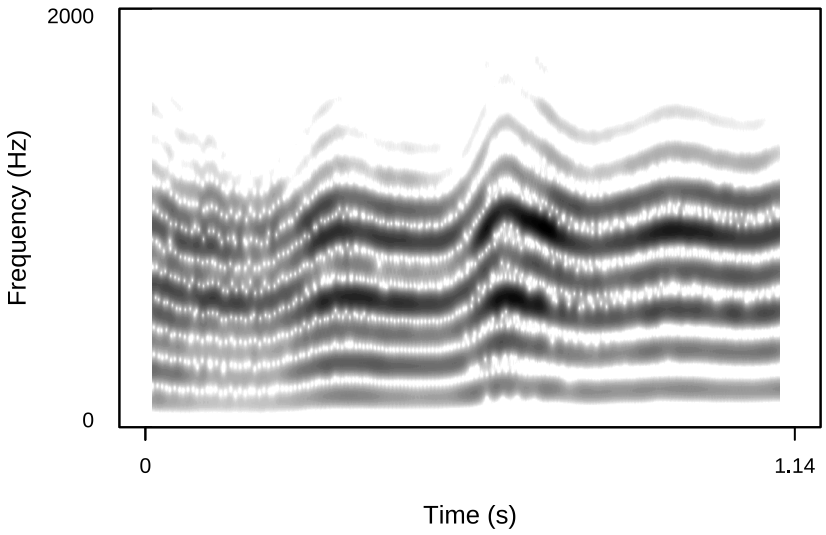
Harmonic	Wave-length	Frequency
0	686 cm	50 Hz
1	1372 cm	100 Hz
2	2058 cm	150 Hz
3	2744 cm	200 Hz
4	3430 cm	250 Hz

Harmonic	Wave-length	Frequency
0	343 cm	100 Hz
1	686 cm	200 Hz
2	1029 cm	300 Hz
3	1372 cm	400 Hz
4	1715 cm	500 Hz

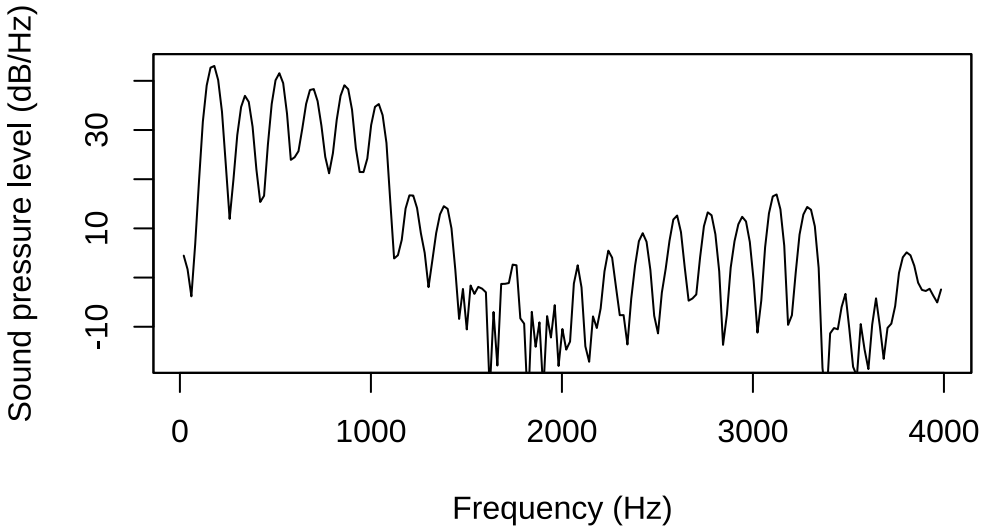
Harmonics



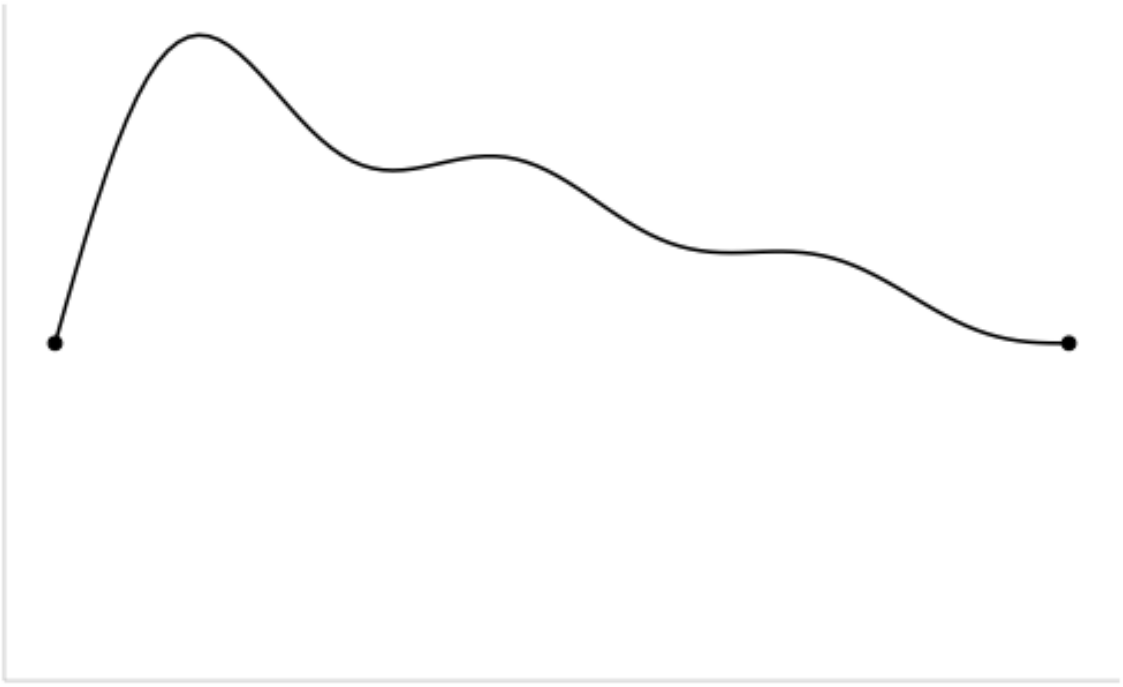
Harmonics



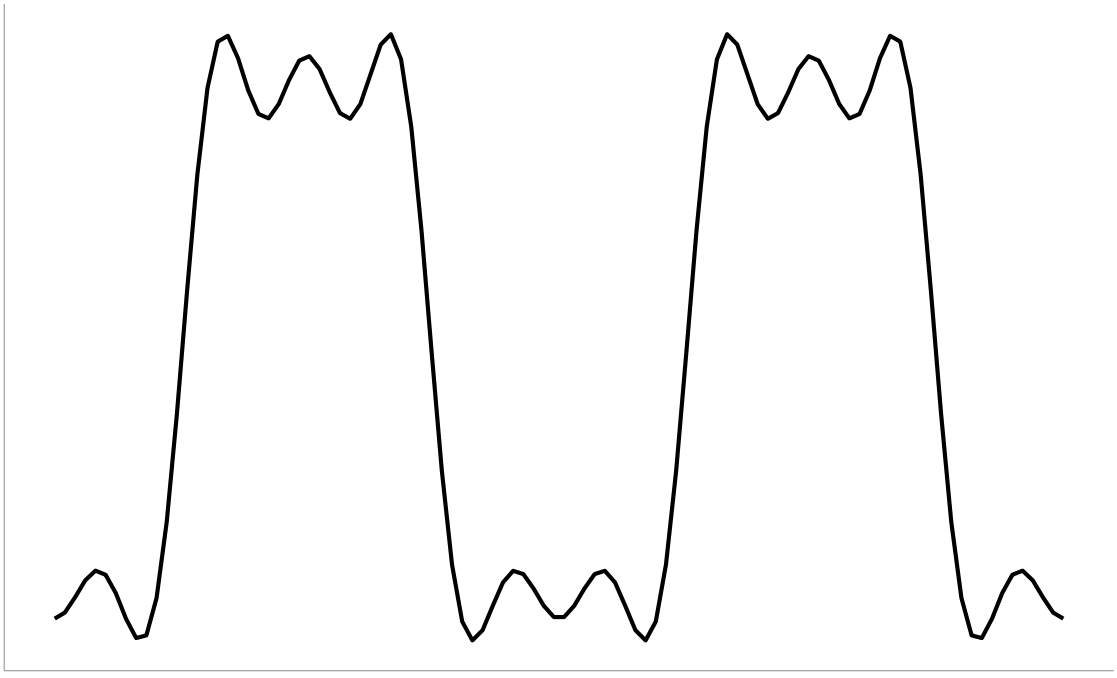
Harmonics



Harmonics

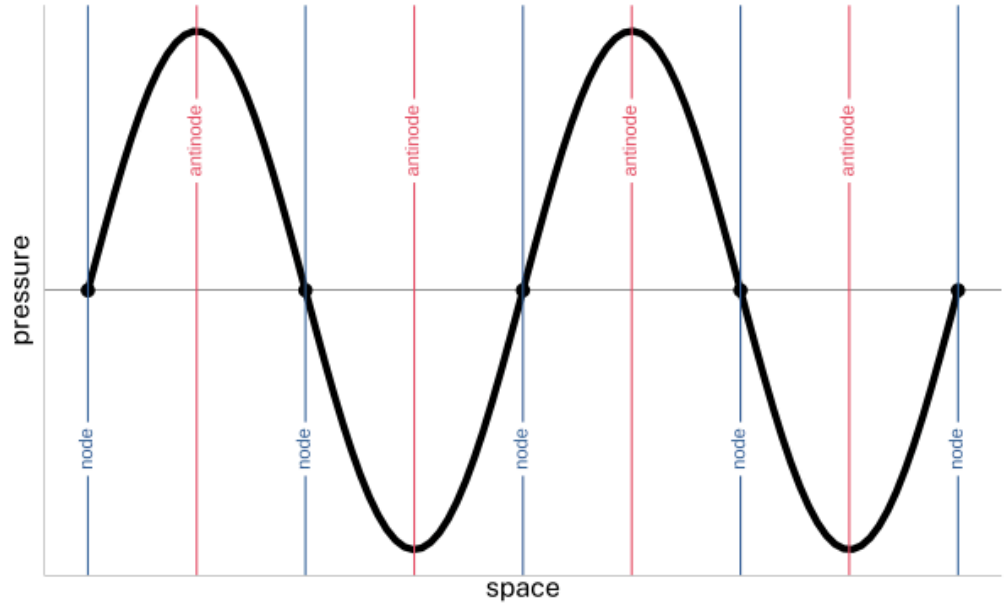


Harmonics



Standing Waves

Nodes and Antinodes

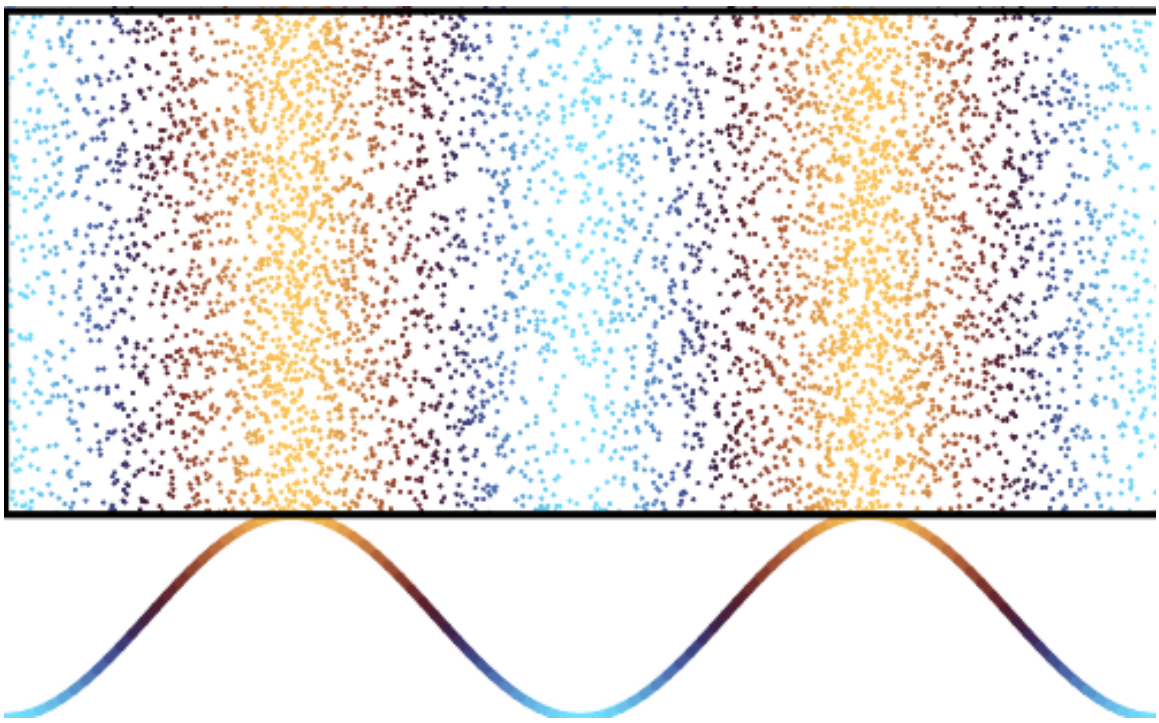


From the source to the filter

- The physical activity of the vocal folds generates the fundamental frequency and the harmonics.
- Then, the air pressure waves propagate through your vocal tract, which introduces *resonances*.

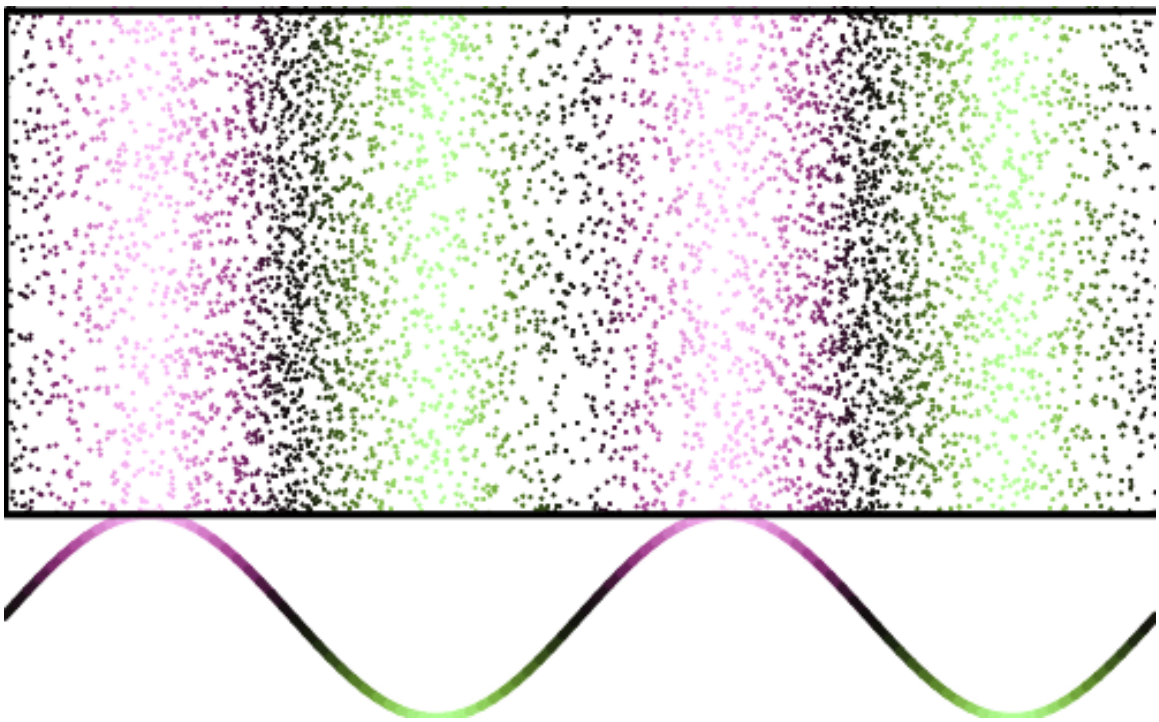
Resonance - From the Filter

Pressure



Pressure & Displacement

Displacement

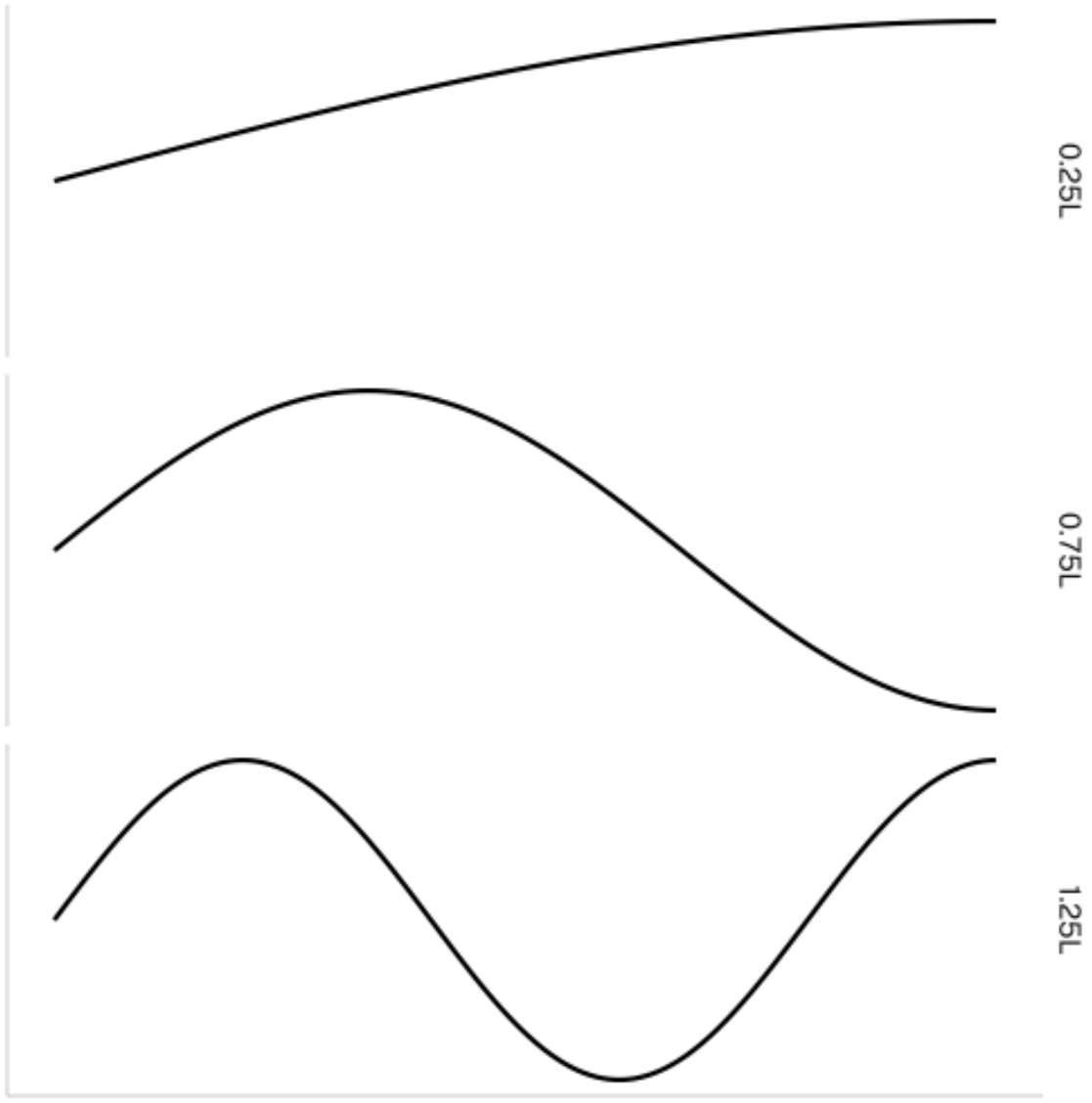


Resonance - From the Filter

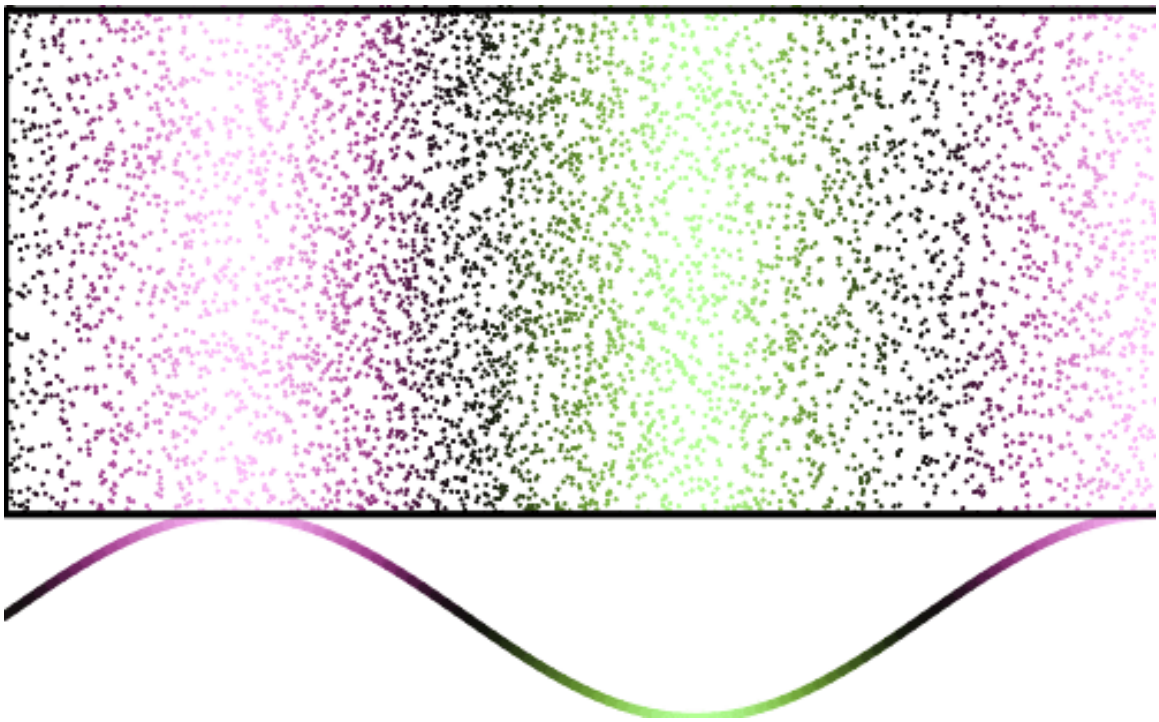
https://www.youtube.com/watch?v=dihQuwrf9yQ

Resonance - From the filter

Wavelengths that put a displacement antinode at the lips are resonant frequencies.


0.25L
0.75L
1.25L

Resonance - From the Filter



Resonance - From the Filter

$$F_n = \frac{(2n-1)c}{4L}$$

$$L = \frac{(2n-1)c}{4F_n}$$

- n = resonant frequency
- c = speed of sound